

A Thematic Layer for CLAUDETTE: Separating the Cost of Multi-Label Annotation from the Effect of Taxonomy Granularity

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Abstract. The CLAUDETTE/UNFAIR-ToS corpus is the reference benchmark for detecting unfair clauses in online terms of service, but it labels unfairness sentence by sentence and records nothing about what each clause is about. We add a thematic layer to its 50 contracts: 9,414 sentences, each labelled with one or more of 20 legal themes independently by three trained annotators, and separately by four large language models prompted with the same vocabulary on the same sentences. The layer separates two effects that the reliability literature tends to conflate. The multi-label format costs 0.073 [0.061, 0.087] of chance-corrected agreement, and this cost varies by only 0.012 across four granularities; consolidating 20 themes into 11 recovers 0.067 [0.058, 0.077], a gain that replicates on 17 held-out contracts. Consolidation is constrained, however: the taxonomy that agreement alone would select merges across unfairness strata and loses 13.2% of the downstream signal. A fine-tuned legal encoder outperforms every judge while remaining below the human ceiling. We release the annotations, codebook, protocol and annotation platform.

Keywords. unfair terms of service, CLAUDETTE, multi-label annotation, inter-annotator agreement, LLM-as-a-judge, legal language models, annotation platform

1. Introduction

CLAUDETTE records whether a sentence in a consumer contract is potentially unfair; it records nothing about what the clause is about [1,2]. A limitation clause and an arbitration clause are both unfair, and both are simply unfair. Where clause topics do exist for contracts, the resources are single-annotated [3], unpublished, or reported without chance-corrected agreement [4], so it is not known how reliably the task can be performed at all.

We add a thematic layer to the reference benchmark. All 50 contracts and 9,414 sentences of CLAUDETTE are labelled with one or more of 20 legal themes, independently by three trained annotators, and separately by four large language models prompted with

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the same vocabulary on the same sentences. Individual votes, the arbitration trail and the resulting gold standard are all released.

The layer settles a question that the reliability literature leaves open. Agreement is known to fall when annotators may assign several labels, and to rise when categories are merged [5,6]; what is not known is whether these are one effect or two, since they are seldom measured on the same sentences. Here they are. The multi-label format costs 0.073 [0.061, 0.087] of chance-corrected agreement, and this cost varies by only 0.012 across four granularities — while merging 20 themes into 11 recovers 0.067 [0.058, 0.077], a gain that replicates on 17 contracts held out from the design of the merges. Consolidation is therefore a real lever on reliability, but not an unconstrained one: the taxonomy that agreement alone would select merges clauses across unfairness strata, and loses 13.2% of the downstream unfairness signal to buy 0.012 of extra agreement.

We contribute: (i) a thematic layer over CLAUDETTE, multi-label and triply annotated, published with its disagreements and with the open-source platform used to produce it; (ii) a separation of the cost of the multi-label *format* from the effect of taxonomy *granularity*, measured on the same sentences and replicated out of sample; and (iii) a legal constraint on consolidation — merges must not cross unfairness strata — together with the price of violating it.

2. Background and Related Work

Automatic detection of potentially unfair clauses in terms of service is by now well established: CLAUDETTE and its successors detect and explain unfairness in English and across languages [1,2,7,8,9,10], and the task is part of standard legal benchmarks [11]. Compact domain-specific encoders remain competitive with, and often better than, general-purpose large models on clause classification [12,13,14,15]. Language models are increasingly used as semantic annotators of legal text, with documented gains in cost and documented risks of anchoring and circularity [16,17,18,19,20], and, as judges, with known divergence from human evaluators [21,22,23]. On the measurement side, chance-corrected agreement for set-valued annotation is a mature subject [24,25,26,6,27], and a meta-analysis of annotation studies identifies the number of categories as one of the strongest determinants of agreement [5].

What is missing is a reliable, measured and published thematic layer on the reference benchmark itself. Contract-topic resources adjacent to ours are either single-annotated [3], defined for a different document type [28,29,30], or reported without chance-corrected agreement [4,31]; none places human annotators and language models on the same vocabulary and the same sentences. The consequence is methodological as much as practical: because the multi-label format and the granularity of the taxonomy are almost never varied on the same data, it is not known whether the agreement penalty attributed to multi-label annotation is a property of the format or of the label set it is applied to — a question [5] raises and leaves open.

3. The Thematic Layer and Its Construction

The layer. We annotate all 50 English contracts of CLAUDETTE [1]: 9,414 sentences, each labelled with one primary theme and zero or more secondary themes drawn from

Table 1. Reliability and downstream signal by taxonomy, on all 50 contracts (9,414 sentences, three annotators). The gap between α -MASI and nominal α is the cost of the multi-label format. Differences against T20 are paired by document; 95% confidence intervals from 1,000 document-level bootstrap resamples. Signal loss is the relative drop in leave-one-document-out average precision of $P(\text{unfair} \mid \text{theme combination})$, scored against the CLAUDETTE unfairness labels, which are never projected. T10 merges across unfairness strata; T14 and T11 do not.

Taxonomy	Classes	α -MASI	Nominal α	Δ vs. T20 [95% CI]	Signal loss
T20	20	0.658	0.732	—	—
T14	14	0.693	0.768	+0.035 [0.030, 0.042]	6.4%
T11	11	0.725	0.794	+0.067 [0.058, 0.077]	5.9%
T10	10	0.737	0.800	+0.078 [0.068, 0.090]	13.2%

a 20-theme legal vocabulary (T20) covering the recurring subject matter of online terms of service — account and permitted use, content and intellectual property, liability, warranty, termination, dispute resolution, personal data, fees, unilateral modification, and third-party services, among others. Every sentence was labelled independently by each of three trained annotators working from a written codebook, and, separately, by four large language models prompted with the same vocabulary on the same sentences. Individual votes are retained rather than collapsed into a consensus, so that disagreement remains available for analysis [32,33,34].

Construction. Annotators post-edited model pre-annotations rather than labelling from a blank slate; this speeds the task and anchors it, and we treat the resulting anchoring as a limitation to be bounded rather than a problem to be assumed away (Section 5). Annotators could not see one another’s work. The gold standard is built by a three-tier cascade: 46.8% of sentences are unanimous, 48.3% carry a majority of at least two of three, and the remaining 4.9% (462 sentences) were arbitrated by a human committee. That last share is not overhead but a measurement: it is the fraction of the corpus on which three trained annotators, given the same codebook, cannot converge. Crucially, *model outputs are never parties to the gold*: resolution is strictly inter-annotator, and the judges are held out as an object of evaluation rather than a contributor to the reference [23].

Measurement. We report Krippendorff’s α under two distances [24,26]: the MASI distance for set-valued labels [25], and the nominal distance on primary themes alone. The gap between them is what multi-label annotation costs in reliability [6]. Coarser taxonomies (T14, T11, T10) are *projections* of T20 applied at read time: the annotations themselves are never rewritten, so every comparison is paired by construction, and confidence intervals come from bootstrap resampling over documents rather than sentences. The merges were designed on 33 contracts and validated on the 17 held out from that design. Finally, the human ceiling is estimated leave-one-annotator-out, each annotator being scored against a gold to which they did not contribute [35].

4. Results

The cost of the multi-label format. Scoring secondary themes lowers agreement from a nominal α of 0.732 to an α -MASI of 0.658: the multi-label format costs 0.073 [0.061,

Table 2. Agreement with the frozen gold standard, for all three kinds of system, measured by the same procedure. Human agreement is leave-one-annotator-out, so that each annotator is scored against a gold to which they did not contribute. LLM judges are evaluated on stored pre-annotations. Reported, not demonstrated: see Section 4.

System	Setting	κ	Accuracy
Human annotators	leave-one-annotator-out	0.859	0.871
Best LLM judge	stored pre-annotations, T20	0.581	0.601
Legal-BERT, fine-tuned	T20, 5-fold by document	0.695	—
Legal-BERT, fine-tuned	T11, 5-fold by document	0.720	—

0.087] of chance-corrected agreement. Across the four granularities the penalty stays within a narrow band, from 0.063 at 10 classes to 0.075 at 14, and it does not decrease monotonically as categories are merged (Table 1). Its whole range, 0.012, is five times smaller than the agreement gain that consolidation yields below. The two effects that the literature tends to conflate are therefore of different orders, and can be measured separately on the same sentences.

The effect of granularity. Consolidating the taxonomy recovers agreement of the same order as the format costs. Merging T20 into T11 raises α -MASI by 0.067 [0.058, 0.077], carrying the layer across the 0.667 threshold conventionally used for tentative conclusions [25] — a threshold the annotation taxonomy itself does not reach. Two properties make this more than a mechanical artefact of coarser categories. First, the gain *replicates out of sample*: the 17 contracts held out from the design of the merges yield +0.069, against +0.066 on the 33 contracts used to design them. Second, T11 *dominates* the intermediate T14: it gains twice as much agreement (+0.067 against +0.035) while losing *less* downstream signal (5.9% against 6.4%). Granularity is therefore a genuine lever on reliability, and not merely a redistribution of the same disagreements over fewer categories. What the result does not establish is that the merged categories are legally better: it measures how consistently annotators apply them, not whether they carve the domain well.

The legal constraint on consolidation. Consolidation cannot be driven by agreement alone. T10, obtained by merging along measured confusion clusters without regard to the legal status of the clauses, buys a further 0.012 of α -MASI over T11 — and loses 13.2% of the unfairness signal instead of 5.9%, a yield of 0.59 points of agreement per unit of signal against 1.13 for T11. The merge responsible is one that crosses unfairness strata, placing clauses that the Directive treats differently [36,37] into a single category. A purely statistical consolidation selects exactly this taxonomy; the legal constraint rules it out. Reliability gains and downstream utility are thus not interchangeable, and the boundary between them is drawn by law, not by data.

Reported: the human ceiling, the judges and a fine-tuned encoder. Against the frozen gold, trained annotators reach $\kappa = 0.859$ in leave-one-annotator-out, the best of the four LLM judges reaches 0.581, and a fine-tuned Legal-BERT reaches 0.720 on T11 (Table 2). We report these figures rather than demonstrate them: the encoder is a single model with early stopping on the test fold, and the judges are scored on stored pre-annotations. Their ordering is nonetheless unambiguous — a compact, domain-specific encoder outperforms every judge while remaining well below the human ceiling.

5. Discussion and Limitations

Two consequences follow for annotation practice. First, the reliability cost of multi-label annotation need not be paid by abandoning multi-label annotation: it can be recovered by consolidating the taxonomy, which preserves the ability to record that a clause does two things at once. Second, that consolidation cannot be delegated to the agreement signal. Merging along measured confusion clusters is what a data-driven procedure would do, and on this corpus it selects precisely the taxonomy that collapses the legal distinction the downstream task depends on. The criterion has to come from outside the annotation statistics — here, from the strata of unfairness that consumer protection law itself draws [36,37]. For the evaluation of language models, the gap between the human ceiling and the best judge (0.859 against 0.581) is large enough that judges are not substitutable for trained annotators on this task, in line with findings on other legal annotation tasks [23,22]; a compact domain-specific encoder [12] closes much, though not all, of that gap, which is consistent with the specialization results of [13].

Limitations. Pre-annotations were post-edited rather than produced from scratch, and annotator divergence from the suggested label (38.5–48.6% of sentences) bounds the anchoring risk [20] without eliminating it. The layer covers one legal system and one language. The judges are scored on stored pre-annotations, so the benchmark measures a historical output rather than the current state of the art. The fine-tuned encoder is a single model with early stopping on the test fold, making its reported agreement mildly optimistic. Finally, the aggregated gold carries only 1.04 labels per sentence, so a multi-label score computed on it would not measure the difficulty of the task as annotated; the multi-label cost reported here is measured on individual votes, where the secondary themes survive.

6. Conclusion and Availability

A thematic layer over CLAUDETTE makes it possible to ask what a clause is about, and not only whether it is unfair. Built as a multi-label annotation by three independent trained annotators, it also makes the reliability of that question measurable: the cost of the multi-label format and the effect of taxonomy granularity are separable, of different orders, and the consolidation that agreement alone would favour is ruled out by a legal constraint that no statistical criterion expresses.

Availability. We release the thematic layer with its individual annotator votes, the arbitration trail and the frozen gold standard; the annotation codebook and protocol; the frozen taxonomy projections (T20, T14, T11, T10) with the fingerprint used for every figure in this paper; and the open-source annotation platform on which the layer was produced, which supports browsing, measuring, arbitrating, exporting and extending it under the same protocol [38,39,40]. **[TODO: DOI, licence and repository URL to be fixed before camera-ready.]**

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